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Bettina Ballin

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Work Experience

Self-employed / IT Freelancer in Software Consulting — Remote / Germany, Apr 2025 – Present
- Architecture and optimization of custom software solutions for SMEs
- Administration of web projects and hosting of IT infrastructure

Fraunhofer SIT, Research Associate in Secure Software Engineering — Darmstadt, Nov 2022 – Apr 2025
- Publications in the field of AI and cybersecurity
- Research on anomaly detection systems for German aerospace companies
- Contribution to penetration tests for public institutions and scientific papers for research partners

Software AG, webMethods Consultant — Darmstadt, Feb 2018 – Jun 2020
- IT consultant in agile teams at leading German banks and insurance companies
- Implementation of backend and frontend solutions using the webMethods integration platform

Technical University of Darmstadt, Student Assistant — May 2022 – Sep 2022
- **Tutorial Statistical Machine Learning**: Supervising exercise groups and supporting class activities
- **PTW Institute:** Implementation of energy flexibility tools in Python, development of web apps using Flask and Docker — Oct 2020 – Apr 2022
- **SECUSO Research Group:** Development of Thunderbird add-ons to protect against phishing attacks — Mar 2016 – Dec 2017

Staufenbiel Institute, Event Hostess & Promotion — Rhine-Main Area, Germany, Oct 2014 – Mar 2015
- Recruiting and event organization focused on young university graduates

Education

M.Sc. Technical University of Darmstadt — Oct 2020 – Present
- M.Sc. Computer Science: Grade: 1.3 (excellent)
- Master's Thesis: *Robustness of Membership Inference Attacks on Neural Networks with Label-Only Exposures*
- Focus on AI technologies, minor in Philosophy

B.Sc. Technical University of Darmstadt — Oct 2013 – Dec 2017
- B.Sc. Computer Science: Grade: 1.9 (good)
- Bachelor's Thesis: *Toolbased Detection of Malicious Links in Online-Email-Applications*

Abitur Ulrich-von-Hutten-Gymnasium, Schlüchtern — Aug 2005 – Jul 2013
- General university entrance qualification

Certificates and Awards

- T.I.S.P. TeleTrust Information Security Professional (2023)
- Germany Scholarship [Deutschlandstipendium] (2020)
- webMethods 9.9 Certified ESB Developer (2018)

Skills and Competencies

- **Programming Languages:** Java, Python, C/C++, C#, SQL, XML, LaTeX
- **Web Development:** HTML, CSS, JavaScript, jQuery, Angular, Flask
- **Technologies:** Docker, Android Studio, Eclipse, Visual Studio Code, GitHub, JIRA, Confluence, Microsoft Office
- **Databases:** Oracle, MongoDB
- **AI & Machine Learning:** Deep Learning, TensorFlow, PyTorch, NLP, Computer Vision, Anomaly Detection
- **IT Security:** Cybersecurity, Penetration Testing, Vulnerability Analysis
- **Operating Systems:** Linux, Windows

Language Skills

- German: Native
- English: Fluent (professional proficiency)
- French, Spanish: Intermediate
- Chinese: Basic

Additional Information

- Mentored in the first-year orientation week at TU Darmstadt (Winter Semester 2016)
- Interests: Piano, logic puzzles, yoga, bouldering
- Strengths: Quick comprehension, strong interest in technical and mathematical challenges, team spirit and communication